



**CARIBBEAN EXAMINATIONS COUNCIL  
ADVANCED PROFICIENCY EXAMINATION**

**ENVIRONMENTAL SCIENCE**

**UNIT 1: ECOLOGY, PEOPLE AND NATURAL RESOURCE USE**

**PAPER 02**

*2 hours 30 minutes*

**08 MAY 2007 (p.m.)**

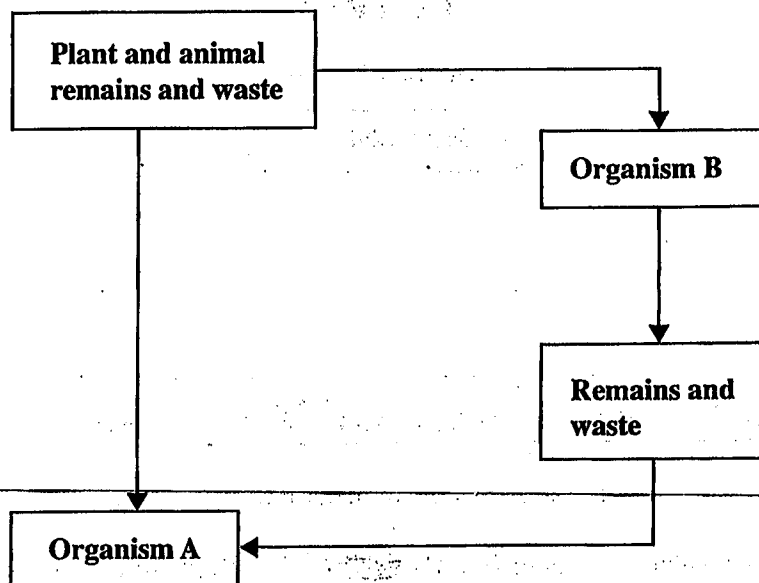
**READ THE FOLLOWING INSTRUCTIONS CAREFULLY**

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Answer **SIX** questions, **TWO** from **EACH** Module.
4. Write your answers in the answer booklet provided.

## MODULE 1

Answer ANY TWO questions.

1. Figure 1 shows a feeding relationship and two groups of organisms, A and B, that exist in ecosystems.



**Figure 1. Feeding relationship in an ecosystem**

- (a) State the name of EACH of the TWO groups of organisms, A and B. Give ONE example of an organism in EACH group. [ 4 marks]
- (b) Explain how the role of EACH group of organisms is important to the functioning of the ecosystem. [ 8 marks]
- (c) Explain how human activity may disrupt the feeding relationship illustrated in Figure 1. Include TWO ways in your response. [ 8 marks]

**Total 20 marks**

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2. (a) What is meant by
- (i) ecological succession? [ 2 marks]
  - (ii) a climax community? [ 2 marks]
- (b) Outline THREE differences between 'primary ecological succession' and 'secondary ecological succession'. [ 6 marks]
- (c) The results of a sampling exercise to determine the size of the fish population in three streams, A, B and C, are presented in Table 1.

**TABLE 1: SAMPLES OF FISH POPULATIONS IN STREAMS A, B AND C**

| Species of Fish                         | Number of Individuals in Stream |             |             |
|-----------------------------------------|---------------------------------|-------------|-------------|
|                                         | A                               | B           | C           |
| <i>Crenicichla saxatilis</i> (sunfish)  | 48                              | 3           | 35          |
| <i>Cichlasoma bimaculatum</i> (patwa)   | 10                              | 8           | 6           |
| <i>Cichlasoma severum</i> (river patwa) | 13                              | 32          | 15          |
| <i>Cichla ocellaris</i> (lukanani)      | 7                               | 2           | 12          |
| <b>Total number of individuals</b>      | <b>78</b>                       | <b>45</b>   | <b>68</b>   |
| <b>Species diversity, (D)</b>           |                                 | <b>1.88</b> | <b>2.92</b> |

- (i) The species diversity for Streams B and C are indicated in Table 1. Calculate the species diversity for Stream A.

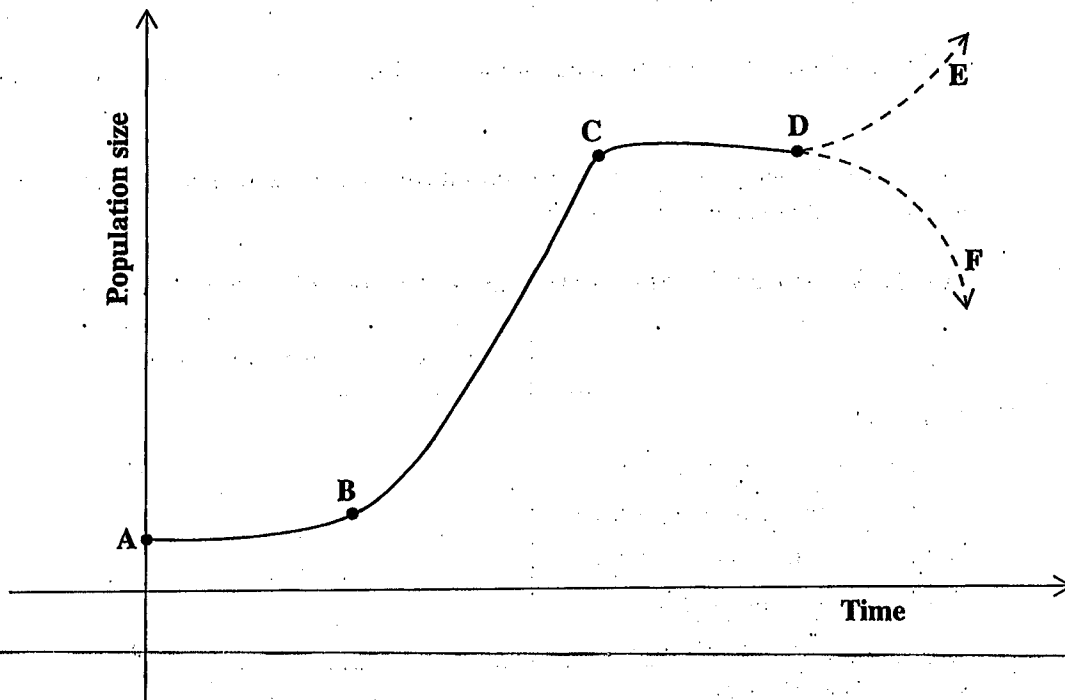
$$D = \frac{N(N-1)}{\sum n(n-1)}$$

[ 4 marks]

- (ii) Rank the THREE streams in decreasing order of species diversity. [ 2 marks]
- (iii) Which of the three streams, A, B or C, is the most stable? [ 1 mark ]
- (iv) Justify your answer to 2 (c) (iii) above. [ 3 marks]

**Total 20 marks**

3. Figure 2 shows a typical population growth curve, with different growth stages represented by the intervals AB, BC and CD.



**Figure 2. Typical population growth curve**

- (a) State the name of ANY ONE of the growth stages, AB, BC or CD, shown in Figure 2. [ 1 mark ]
- (b) Outline the features of EACH growth stage, AB, BC and CD, shown in Figure 2. [ 9 marks ]
- (c) Explain how environmental resistance influences the shape of the curve from A to D in Figure 2. [ 6 marks ]
- (d) Beyond point D, the population may grow along the interval DE or DF, as indicated by the broken portions of the curve in Figure 2.
- Suggest ONE environmental condition in the ecosystem that would cause growth along the interval
- (i) DE [ 2 marks ]
- (ii) DF. [ 2 marks ]

**Total 20 marks**

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## MODULE 2

Answer ANY TWO questions.

4. Table 2 shows selected statistics for Countries A and B.

**TABLE 2: SELECTED STATISTICS FOR COUNTRIES A AND B**

|           | Size of<br>Population<br>(mid 2003) | Size of<br>Population<br>(mid 2004) | Doubling time<br>(years) | Per capita<br>GNP (US\$) |
|-----------|-------------------------------------|-------------------------------------|--------------------------|--------------------------|
| Country A | 268,000                             | 269,340                             | 140                      | 7500                     |
| Country B | 8,056,000                           | 8,205,000                           | —                        | 400                      |

- (a) Calculate the doubling time for the population of Country B. [ 4 marks]
- (b) Using the result from 4 (a) above and the information in Table 2, evaluate whether both countries need to implement population control measures. [ 6 marks]
- 
- (c) (i) State TWO factors that affect the population growth rate of a country. [ 2 marks]
- (ii) Explain how EACH of the factors stated in (c) (i) above affects human population growth rate. [ 8 marks]

**Total 20 marks**

5. For Country Y, the following statistics were recorded for 2004.

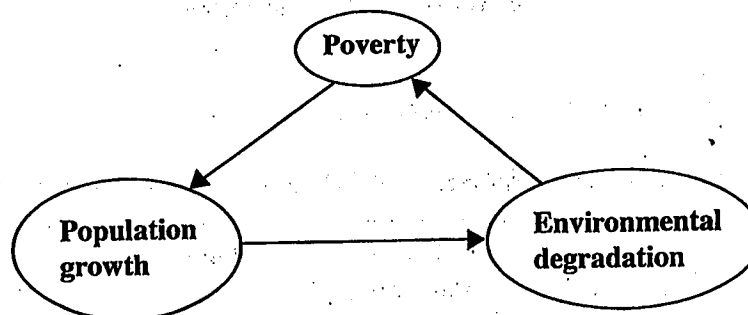
|                       |   |                        |
|-----------------------|---|------------------------|
| Infant Mortality Rate | — | 108                    |
| Total fertility rate  | — | 5.7                    |
| Rate of Deforestation | — | 1.250 million hectares |

- (a) Define 'total fertility rate'. [ 2 marks]
- (b) Compare the infant mortality rate and the total fertility rate of Country Y with average rates in Caribbean countries. [ 7 marks]
- (c) Explain how infant mortality rate and fertility rate impact on the rate of growth of a population. [ 5 marks]
- (d) Describe the relationship between the rate of growth of the population and the rate of deforestation. [ 6 marks]

**Total 20 marks**

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6. Figure 3 illustrates a proposed relationship between poverty, population growth and environmental degradation.



**Figure 3. Cycle of poverty**

- (a) (i) Define 'poverty'. [ 2 marks]
- (ii) Justify the relationship in the 'cycle of poverty' illustrated in Figure 3. [ 5 marks]
- (b) (i) State THREE social impacts of poverty. [ 3 marks]
- (ii) What is meant by 'sustainable development'? [ 2 marks]
- (iii) "Finding solutions to the problems of poverty is necessary if a country is to achieve sustainable development".
- Evaluate the statement above. In your evaluation, present at least TWO reasons to support your position. [ 8 marks]

**Total 20 marks**

### **MODULE 3**

**Answer ANY TWO questions.**

7. (a) Explain why water is considered a 'natural resource'. [ 3 marks]
- (b) Explain how water is important as a natural resource in a **named** Caribbean territory. Include FOUR uses of water in your answer. [ 9 marks]
- (c) Explain how ANY TWO of the measures listed below are important in the management of water resources.
- (i) Education and awareness
- (ii) Legislation
- (iii) Environmental Impact Assessments (EIAs) [ 8 marks]

**Total 20 marks**

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8. (a) State FOUR environmental impacts associated with the use of land resources in a named Caribbean country. [ 4 marks]
- (b) Describe TWO of the environmental impacts listed in 8 (a) above. [ 8 marks]
- (c) Describe TWO measures you would recommend for addressing EACH of the impacts described in 8 (b) above. [ 8 marks]

**Total 20 marks**

9. A major limestone deposit is located in the interior of a Caribbean island. This area is hilly and situated away from regular communication and transportation networks. While it is recognized that this is a major deposit of a natural resource with significant potential, it is not exploited because of many different factors.

- (a) List THREE factors that may impact on the exploitation of the limestone resource mentioned above. [ 3 marks]
- (b) Explain how EACH of the factors listed in 9 (a) could impact on the exploitation of the limestone resource mentioned above. [ 9 marks]
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- (c) Outline FOUR environmental impacts that may result from the exploitation of the limestone resource. [ 8 marks]

**Total 20 marks**

**END OF TEST**